

MITHILESH KUMAR UPADHYAY

AI Solutions Architect | Tech Manager IT & Software Operations

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PROFESSIONAL SUMMARY

Dynamic IT Leader and AI Solutions Architect with over 15 years of robust software industry experience, featuring a stellar 12-year progressive career at **Samsung India Electronics** (advancing through 4 rapid promotions from Engineer to Manager). Expertly capitalizes on a deep foundation in enterprise software engineering (C#, .NET, SQL Server, PHP) to architect, deploy, and scale advanced **Generative AI systems, Multi-Agent Workflows, and LLM-powered enterprise solutions**. Proven track record of leading high-performing technical teams, modernizing legacy systems, and delivering high-scale digital solutions that bridge traditional corporate infrastructure with state-of-the-art AI. Open to leadership roles and elite consulting/freelance engagements.

CORE AI & TECHNICAL EXPERTISE

- **Generative AI & LLMs:** Lang Chain, Llama, Llama 3, Google Gemini, Grog, Grog Cloud GCP.
 - **AI Architecture:** Multi-Agent Workflows, Autonomous AI Agents, RAG (Retrieval-Augmented Generation), Prompt Engineering.
 - **Enterprise Development:** ASP.NET (C#), MVC, PHP, Node.js, Web API, WCF, Windows Services.
 - **Data & Database Engineering:** SQL Server (2012–2022), Oracle 11g, SSRS, SSIS, T-SQL Optimization.
 - **Cloud & Devops:** GCP (Google Cloud Platform), Git, TFS, Agile/Scrum Methodologies.
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CHRONOLOGICAL PROFESSIONAL EXPERIENCE

SAMSUNG INDIA ELECTRONICS PVT. LTD. | Noida, India

Manager – IT / Software Operations & AI Integration | *September 2013 – Present*

Progressive 12-year career with 4 rapid promotion cycles driven by technical excellence and leadership impact.

- **AI Modernization Strategy:** Direct the digital transformation roadmap, spearheading the integration of **Generative AI models and Lang Chain workflows** into Samsung's legacy manufacturing software ecosystem.
- **Strategic Team Leadership:** Lead an agile, cross-functional engineering team. Architecting modern AI solutions while ensuring 100% uptime for high-scale enterprise operations.
- **Agentic Workflows:** Designing autonomous **AI Agents** to automate complex data analysis, internal reporting, and root-cause monitoring within the production floor environment.
- **Assistant Manager [2018 – 2025]:**
 - **Project Oversight & SDLC:** Managed full-lifecycle software delivery for mid-to-large scale internal applications. Successfully aligned technical roadmaps with corporate business goals.
 - **Process Optimization:** Standardized enterprise version control (Git/TFS) and continuous improvement metrics across multi-tiered .NET and PHP software architectures.
- **Senior Engineer [2015 – 2018]:**
 - **Database Architecture:** Engineered high-load SQL Server database systems. Built high-performance backend C# services to handle real-time enterprise data synchronization.
 - **Tier-3 Technical Escalation:** Served as the core subject matter expert resolving critical, high-impact systemic bugs on the manufacturing production lines.
- **Engineer [2013 – 2015]:**
 - Developed core web-based infrastructure tools utilizing PHP and ASP.NET. Authored highly optimized stored procedures and T-SQL structures.

EXCEL SOFT TECHNOLOGY PVT. LTD. | Noida, India

Software Engineer | *February 2012 – September 2013*

- Architected enterprise web components using ASP.NET, C#, and ADO.NET.
- Engineered high-performing reporting metrics using Google Charts and SSRS to provide executive leadership with real-time operational visibility.

BINARY SOFT ZONE PVT. LTD. | Noida, India

Software Engineer / Trainee | *October 2010 – February 2012*

- Contributed to frontend and backend application cycles using PHP, C#, and SQL Server.
 - Built and deployed custom DLLs, window forms, and port programming scripts.
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AI PROJECT HIGHLIGHTS

1. Enterprise Multi-Agent Data Analyst (Lang Graph & Grog)

- **Description:** Built a multi-agent system using **Lang Graph** and **Grog Cloud** to automate internal data analytics and reporting.
- **How it Works:** The system orchestrates two specialized agents: a *Data Extractor Agent* that converts natural language into optimized SQL queries via internal APIs, and a *Business Analyst Agent* that interprets the database output.
- **Impact:** It delivers instant, production-grade business insights and charts to end-users without requiring manual data intervention.

2. Multimodal Manufacturing Inspection & Inventory Agent (Gemini API & GCP)

- **Description:** Developed an autonomous multimodal agent deployed on **GCP** utilizing the **Google Gemini API** to process both text inputs and factory floor imagery.
- **How it Works:** Users upload real-time images of hardware inventory or equipment parts; the agent analyses the visual components, detects anomalies, and cross-references an internal parts database.
- **Impact:** It automatically generates structural health logs and triggers restocking API alerts, cutting down manual inspection time significantly.

3. Local Privacy-First Knowledge Base (Lang Chain, Llama & Llama 3)

- **Description:** Engineered a highly secure, local Retrieval-Augmented Generation (RAG) system using **Lang Chain** and **Llama** running **Llama 3**.
 - **How it Works:** The application ingests massive volumes of unstructured corporate documents (PDFs, manuals, API documentation), embeds them into a local vector database, and queries them locally.
 - **Impact:** This ensures absolute data privacy by preventing sensitive enterprise information from leaking to public internet-facing LLM APIs while providing instant semantic search.
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